Uncertainty-Aware Exchange Consistency Learning for Robust Semi-Supervised Remote Sensing Change Detection

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***Abstract*—To alleviate the reliance on large volumes of labeled remote sensing images (RSIs), semi-supervised learning has emerged as a promising paradigm for change detection (CD) applications. Due to the diversity of RSI acquisition conditions, pseudo-changes arising from radiometric discrepancies and seasonal variations can limit the applicability of CD methods in complex scenarios, especially in semi-supervised settings with abundant unlabeled data. However, existing semi-supervised CD (SSCD) methods often focus on the discrepancy between labeled and unlabeled samples rather than the internal diversity within bitemporal pairs, thus struggling to cope with these pseudo-change issues. In this work, we propose a robust consistency regularization framework for SSCD that establishes a unified and comprehensive perturbation space via intertemporal, intratemporal, and feature-level perturbations, improving network adaptability to complex scenarios. Specifically, a novel and effective data perturbation approach, SwapMix, is proposed to explicitly model cross-temporal discrepancies and to embed intertemporal perturbations tailored for SSCD tasks. Subsequently, we design an exchange consistency learning (ECL) strategy to enforce prediction consistency between the multibranch perturbed inputs for labeled and unlabeled data, encouraging the network to learn feature representations that are robust to radiometric inconsistencies and seasonal variations. To stabilize semi-supervised learning, we design an uncertainty-driven perturbation-specific modulation (UPSM) mechanism that quantifies the reliability of multiple consistency-learning streams and adaptively modulates the training process, thereby attenuating the influence of inaccurate supervisory signals. Consequently, our proposed uncertainty-aware ECL (UA-ECL) approach shows superior performance compared with state-of-the-art methods on the learning, vision, and remote sensing change detection (LEVIR-CD), season-varying change detection (SVCD), and Sun Yat-sen University change detection (SYSU-CD) datasets.**

***Index Terms*—Change detection (CD), consistency regularization, intertemporal perturbation, semi-supervised learning, uncertainty estimation.**

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Nomenclature

D_l, D_u	Labeled and unlabeled datasets of the training set.
$(\mathbf{x}_l^{t_0}, \mathbf{x}_l^{t_1})$	Prechange and postchange images of labeled sample pair.
$\mathbf{y}_l$	Ground truth of labeled sample pair.
$(\mathbf{x}_u^{t_0}, \mathbf{x}_u^{t_1})$	Prechange and postchange images of unlabeled sample pair.
N_l, N_u	Numbers of labeled and unlabeled sample pairs.
$\mathbf{f}_{t_0}, \mathbf{f}_{t_1}$	Deep features of prechange and postchange images.
$\mathcal{F}(\cdot), \mathcal{G}(\cdot)$	Encoder and the decoder of change detection (CD) network.
B_l, B_u	Mini-batch of labeled and unlabeled data.
$\mathcal{A}^w, \mathcal{A}^s$	Weak and strong perturbation operations.
$\mathcal{E}(\cdot)$	SwapMix operation.
$(\mathbf{x}_{uw}^{t_0}, \mathbf{x}_{uw}^{t_1})$	Unlabeled sample pair after weak perturbations.
$(\mathbf{x}_{us}^{t_0}, \mathbf{x}_{us}^{t_1})$	Unlabeled sample pair after intratemporalstrong perturbations.
$(\mathbf{x}_{ue}^{t_0}, \mathbf{x}_{ue}^{t_1})$	Unlabeled sample pair after intertemporal strong perturbations.
$\mathbf{p}_u^w$	Predictions of weakly perturbed stream.
M	High-confidence mask from weakly perturbed predictions
$\mathbf{p}_u^e, \mathbf{p}_u^s, \mathbf{p}_u^{fp}$	Predictions from the intertemporal, intratemporal, and feature-level perturbed stream of unlabeled data.
$\mathbf{p}_l^w, \mathbf{p}_l^e$	Predictions from the weakly and intertemporal perturbed stream of labeled data.
$\mathbf{u}^*$	A unified representation of pixel-level uncertainty map.

I. Introduction

With the rapid development of Earth observation technology, numerous advanced remote sensing (RS) satellites and autonomous aerial vehicles can provide continuous multimodal and multiresolution data [1], forming a solid foundation for large-scale, long-term environmental monitoring [2], [3], [4]. As a result, CD, which identifies changes in land cover by comparing RS images (RSIs) acquired at different times, has become a significant research topic [5],

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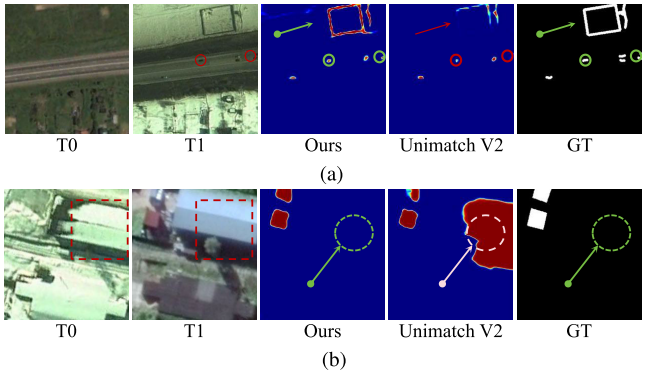


Fig. 1. Examples of challenges to CD applications. (a) Diverse change patterns with different spatial structures. (b) Pseudo-changes caused by radiometric differences and seasonal changes.

[6], [7], and is widely applied in various domains, such as urbanization [8], [9], deforestation [10], disaster assessment [11] and nonagriculturalization [12], [13]. Due to their advantages in automatic multilevel semantic feature extraction, deep-learning-based CD methods [14] have gradually surpassed conventional approaches based on handcrafted features [15], [16], [17] and have become the mainstream technique. However, the advanced fully supervised CD methods based on convolutional neural network (CNN) [18], [19], [20], Transformer [21], [22], and Mamba [23], [24] rely heavily on extensive, dense pixel-level annotations, which impose a substantial annotation burden and significantly hamper their large-scale application in real scenarios. Therefore, semi-supervised learning has been increasingly adopted in CD to alleviate the labeling burden by exploiting the potential of abundant unlabeled samples, which contain diverse change information and are cost-effectively collected by RS platforms [25].

Existing semi-supervised CD (SSCD) methods can be categorized into three paradigms. Adversarial-based methods [26], [27] implicitly align the feature distributions of labeled and unlabeled data through adversarial competition. Pseudo-labeling-based self-training strategies [28], [29], [30] explicitly construct pseudo-labels for unlabeled samples to facilitate subsequent model retraining. Consistency regularization approaches [31], [32], [33] mitigate confirmation bias [34] by enforcing consistent predictions for unlabeled RSI pairs under various perturbations. Through these semi-supervised learning strategies, the CD network achieves performance improvements with limited manual annotations. However, the discrepancy between bitemporal inputs from unlabeled data inevitably increases the difficulty of semi-supervised learning. As shown in Fig. 1, real-world change objects exhibit enormous variations in spatial scale and category, ranging from small cars to extensive artificial structures. In addition, pseudo-changes resulting from radiometric inconsistencies and seasonal variations alter the visual appearance of land surfaces and strongly confound the identification of interested land-cover transitions [3]. To address these challenges, weak-to-strong consistency CD methods promote stable predictions across perturbations of varying strengths, thereby enhancing diverse feature representation learning and mitigating the negative impact of nonsemantic discrepancies on CD. Their performance relies on the strong augmentation techniques and the generation of reliable pseudo-labels from weakly perturbed samples. However, arising from complex physical imaging processes and spatiotemporal interactions, disturbances such as radiometric variations and seasonal vegetation dynamics exceed the modeling capacity of conventional visual augmentations. Moreover, existing augmentation strategies typically operate on individual images, ignoring intertemporal perturbations that can facilitate interactions among heterogeneous images in a siamese network [35]. In addition, the significant discrepancy between labeled and unlabeled data across diverse RS scenarios makes it challenging to obtain reliable pseudo-labels, leading to unstable semi-supervised training.

To address these issues, we propose a novel uncertainty-aware exchange consistency learning (UA-ECL) strategy that comprises a multilevel perturbation strategy, a multi-branch weak-to-strong consistency learning framework, and an uncertainty-driven perturbation-specific modulation (UPSM) mechanism. First, to broaden the perturbation space, we integrate intertemporal, intratemporal, and feature-level perturbation strategies within a unified framework, organizing them into parallel streams to facilitate consistency learning for robust feature representations. Specifically, we develop a tailored data perturbation strategy, SwapMix, for dual-temporal inputs in CD tasks, aiming to exploit cross-temporal interactions and enhance the robustness of CD networks to nonsemantic variations in SSCD. Instead of applying perturbations to each temporal image independently, SwapMix cuts and swaps patches across corresponding regions between bitemporal pairs, thereby constructing more challenging cross-temporal consistency learning scenarios that exploit inherent bitemporal radiometric discrepancies. In addition, SwapMix restructures spatial object patterns by mixing cross-temporal patches, yielding more diverse and challenging bitemporal instances for SSCD training. Second, we propose an exchange consistency learning (ECL) strategy that jointly optimizes labeled and unlabeled data by enforcing prediction consistency across multiple branches of perturbed inputs. By incorporating the multilevel perturbations, ECL drives the network to extract semantic change cues while suppressing irrelevant environmental context, thereby yielding discriminative and robust feature representations. Third, to enhance the stability of training on unlabeled data, we designed a UPSM mechanism to independently evaluate the reliability of supervisory signals for each perturbation stream and adaptively modulate the network optimization. Specifically, UPSM constructs perturbation-specific uncertainty maps to suppress regions sensitive to different perturbations, which typically correspond to ambiguous or pseudo-change areas arising from nonsemantic variations. By integrating these components into a unified multilevel perturbation framework, UA-ECL achieves robust CD performance and can effectively handle the challenges posed by diverse and complex RS scenarios. The main contributions of this article are summarized as follows.

1) We propose a novel SSCD method that integrates intertemporal, intratemporal, and feature-level

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perturbations for labeled and unlabeled data into parallel streams within a unified multibranch consistency-learning framework, enabling stable and robust CD in the extended perturbation space.

2) To mitigate the negative influence of pseudo-changes induced by radiometric and seasonal discrepancies, we propose an innovative intertemporal perturbation strategy tailored for bitemporal inputs to inject temporal disturbances and guide the network to focus on semantic change content.
3) We design a UPSM mechanism that enables pixel-wise reliability estimation and uncertainty-aware consistency optimization to alleviate the adverse effects of perturbation-sensitive pseudo-changes and semantically ambiguous areas on unsupervised learning.
4) Extensive experiments on the learning, vision, and remote sensing change detection (LEVIR-CD), season-varying change detection (SVCD), and Sun Yat-sen University change detection (SYSU-CD) benchmarks evaluate the effectiveness of the proposed components and demonstrate that our proposed framework achieves state-of-the-art performance compared with existing SSCD methods.

The rest of this article is organized as follows. Section II reviews works related to this study. Section III details the proposed UA-ECL approach. Section IV presents a comprehensive evaluation and a detailed analysis of UA-ECL. Conclusions and future research are summarized in Section V.

II. Related Works

A. Fully Supervised CD

By training deep neural networks on large-scale, well-labeled datasets [14], fully supervised CD methods benefit from the automatic feature extraction and strong representation capabilities of deep networks, achieving remarkable success. As an early representative, FC-Siam [18] first introduces the fully convolutional network into a siamese architecture to achieve end-to-end CD. To alleviate the loss of spatial detail and semantic inconsistency issues of FC-Siam, Fang et al. [20] introduced a densely connected siamese network with an ensemble channel attention module to refine and aggregate multilevel features. To further eliminate noisy boundaries and improve object-level geometric awareness, Zhang et al. [36] proposed an end-to-end trainable superpixel-enhanced CD network to integrate deep siamese CNNs with task-adaptive superpixel segmentation. With the success of Transformers in capturing long-range dependencies and modeling global context [37], hybrid models [22] have emerged to balance the capture of local details and broader context by combining CNNs with Transformers. Building on this, Zhang et al. [21] proposed a Swin Transformer-based CD architecture to demonstrate the potential of an end-to-end transformer network for CD. To further reduce computational burden, Chen et al. [23] introduced the Mamba architecture, which achieves linear computational complexity and superior efficiency, enabling faster inference and scalable modeling of the long, multitemporal sequences inherent to CD tasks.

However, training an effective model typically requires a large amount of manually annotated data, which significantly limits the feasibility of these advanced methods in real large-scale applications.

B. Semi-Supervised CD

To enhance the applicability of deep learning models across diverse scenarios, SSCD methods aim to facilitate representation learning from abundant unlabeled RSI pairs, thereby reducing reliance on manual annotations. Early adversarial-learning-based methods improve the generalization of CD networks by aligning the feature distribution of labeled and unlabeled samples through adversarial training [38], [39]. By training a discriminator to distinguish between real samples and generated fake pairs, Gong et al. [26] leveraged adversarial learning to exploit the distribution of unlabeled data, thereby improving the CD network's ability to extract discriminative features across diverse data conditions. In addition, Peng et al. [27] employed dual discriminators to align the change maps and entropy maps generated by a UNet++ model from labeled and unlabeled inputs, thereby aligning these feature distributions. However, since the instability and mode collapse in adversarial learning make optimization challenging, an alternative SSCD strategy has been proposed to generate pseudo-labels for unlabeled data [28], [29], [40], [41]. Assuming that unlabeled samples share the same categories as nearby labeled samples, Liu et al. [42] enriched the training set around labeled samples via active learning. To achieve a more flexible SSCD strategy, Shi et al. [43] combined predictions with clustering results from hidden layers to obtain high-confidence pseudo-labels, which are then iteratively optimized alongside labeled data. To improve context correlation in the single-pixel threshold filtering process, Zuo et al. [44] evaluated the reliability of each instance object by linking it to the context's semantic information. Although rigorous noisy pseudo-label filtering strategies have been incorporated into semi-supervised frameworks, the network's inevitable initial errors tend to accumulate and propagate during iterative training. To alleviate this, recent work has explored strategies such as ensembling multiple student models with a teacher model for mutual correction [45], and hybrid CNN and Transformer architectures to capture local and global features in the early training stage [46]. Furthermore, Zhou et al. [47] simulated changes via multiscale masking to enable self-supervised hyperspectral CD without bitemporal data or annotations. However, the manually designed change patterns deviate from real-world distributions, limiting generalization.

To derive robust and discriminative feature representations under limited labeled data, consistency regularization approaches [48], [49] enforce consistent predictions for unlabeled RSI pairs under various perturbation versions by applying appropriate augmentation techniques. By aligning both weakly and strongly perturbed versions, weak-to-strong consistency regularization approaches, including FixMatch [50] and Unimatch [34], have been shown to significantly improve CD performance, thereby motivating the development of CD-specific consistency regularization methods [51], [52], [53]. Gedara Chaminda Bandara and Patel [32] focused on

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perturbation layers suitable for CD and obtained a robust change detector by ensembling multiple types of perturbations applied to the difference of the feature maps rather than operating at the image level. Subsequently, to enhance the physical interpretability of perturbations in CD, Zhang et al. [31] employed class-aware feature alignment and pixelwise prediction alignment, which reduce intraclass feature discrepancies and improve prediction consistency under diverse transformations. To improve feature-level discrimination between semantic and irrelevant changes, Han et al. [54] proposed a coarse-to-fine CD network with a multiscale attention mechanism, thereby progressively refining change representations. Following that, Liu et al. [55] further incorporated consistency constraints at the pixel, region, and image levels to achieve multilevel feature alignment and collaborative optimization. Nonetheless, radiometric differences in bitemporal images introduce significant intraclass variance [56], which impairs the model's ability to perceive diverse change patterns. However, intratemporal augmentation techniques limit the perturbation space of bitemporal data, thereby restricting the model's ability to capture intraclass invariance across diverse radiometric conditions.

Therefore, it is necessary to explore perturbation strategies on bitemporal inputs within a stable consistency regularization framework, fully exploiting the rich information in temporal pairs to improve the model's ability to distinguish true changes from pseudo-changes caused by radiometric and seasonal variations.

C. Uncertainty Estimation in CD

Uncertainty in model predictions is commonly categorized into epistemic uncertainty, arising from the model's insufficient knowledge of the data distribution, and aleatoric uncertainty, stemming from noise in the observational data. Since pseudo-labels inevitably involve both epistemic and aleatoric uncertainties in SSCD tasks, directly applying them as supervision signals can be detrimental to representation learning from unlabeled data. To construct a reliable dataset for retraining, Wang et al. [29] estimated uncertainty by measuring the consistency of predictions across multiple historical checkpoints and used it to select samples with high prediction confidence. Furthermore, Ran et al. [30] identified the most uncertain regions using pixel-wise information entropy and replaced them with reliable content from labeled or unlabeled datasets, allowing dynamic adjustment of pseudo-labels throughout training. Similarly, Shen et al. [33] adopted an uncertainty-aware decoder in a weakly perturbed branch to produce probabilistic uncertainty maps, which they used as weighting factors in the loss function to emphasize reliable samples and mitigate the impact of noisy or uncertain data during training.

Regions with pseudo-change and semantically ambiguous areas typically exhibit higher prediction uncertainty across perturbations and are more susceptible to perturbation-induced variations, leading to unreliable semi-supervised training. Therefore, we generate pixel-level perturbation-specific uncertainty maps from multiple perturbation views and down-weight the contribution of high-uncertainty regions in the consistency loss, thereby further stabilizing the training of the consistency regularization framework.

III. Methodology

A. Multibranch Consistency Learning Framework

As presented in Fig. 2, our proposed UA-ECL framework is based on a weak-to-strong consistency learning paradigm with a supervised branch trained on limited labeled data $D_l = \{(\mathbf{x}_l^{t_0}, \mathbf{x}_l^{t_1}, \mathbf{y}_l)\}_i^{N_l}$ and an unsupervised branch on the abundant unlabeled subset $D_u = \{(\mathbf{x}_u^{t_0}, \mathbf{x}_u^{t_1})\}_i^{N_u}$. Specifically, different perturbation methods are applied to the input data, forming multiple streams for multibranch consistency learning. In addition, a UPSM mechanism is adopted in the unsupervised branch to enable reliable, adaptive training on unlabeled data. We adopt a baseline CD network comprising a siamese encoder for feature extraction and a change decoder for detecting temporal differences. The notations and definitions used in our method are presented in detail in Nomenclature.

1) Overall Architecture of CD Network: The CD network identifies change regions by analyzing the differences in high-dimensional features extracted from bitemporal data. Specifically, the bitemporal RSI pairs are fed into a siamese encoder $\mathcal{F}(\cdot)$ to extract hierarchical deep features $\mathbf{f}_{t_i} \in \mathbb{R}^{\frac{H}{S}\times\frac{W}{S}\times C}$

$$\mathbf{f}_{t_0} = \mathcal{F}\left(\mathbf{x}^{t_0}\right), \mathbf{f}_{t_1} = \mathcal{F}\left(\mathbf{x}^{t_1}\right) \tag{1}$$

where S and C represent the downsampling ratio and the number of feature channels, respectively; H and W denote the height and width of RSI, respectively.

Following that, the difference features $\mathbf{f}_d$ obtained by $\mathbf{f}_{t_0}$ and $\mathbf{f}_{t_1}$ are fed into the decoder $\mathcal{G}(\cdot)$ with softmax function to generate probability predictions $\mathbf{p} \in \mathbb{R}^{H\times W\times 2}$ for CD

$$\mathbf{p} = \mathcal{G}\left(\left|\mathbf{f}_{t_0} - \mathbf{f}_{t_1}\right|\right). \tag{2}$$

2) Multibranch Consistency Regularization Learning for CD: To improve generalization performance with limited labeled data, the consistency regularization framework is jointly trained on the labeled mini-batch B_l and the unlabeled mini-batch B_u by combining a supervised loss with an unsupervised consistency regularization loss. Fig. 3 illustrates a basic paradigm of consistency regularization learning. The supervised branch of the baseline weak-to-strong consistency learning strategy is optimized by the cross-entropy loss between the predictions and the ground truth. For the unsupervised branch, each unlabeled image $\mathbf{x}_u^{t_i}$ is simultaneously perturbed by weak perturbations $\mathcal{A}^w$, and intratemporal strong perturbations $\mathcal{A}^s$

$$\mathbf{x}_{uw}^{t_i} = \mathcal{A}^w\left(\mathbf{x}_u^{t_i}\right), \mathbf{x}_{us}^{t_i} = \mathcal{A}^s\left(\mathcal{A}^w\left(\mathbf{x}_u^{t_i}\right)\right). \tag{3}$$

With strong perturbations, the appearances of $(\mathbf{x}_{us}^{t_0}, \mathbf{x}_{us}^{t_1})$ undergo significant shifts while preserving the underlying semantic invariance, leading to broader distributional diversity. In contrast, the distribution of $(\mathbf{x}_{uw}^{t_0}, \mathbf{x}_{uw}^{t_1})$ with weak perturbations is similar to the original inputs, yielding more reliable $\mathbf{p}_u^w$ that can be used as pseudo-labels for $\mathbf{p}_u^s$ from $(\mathbf{x}_{us}^{t_0}, \mathbf{x}_{us}^{t_1})$

$$\mathbf{M} = \mathbb{1}\left(\max\left(\mathbf{p}_u^w\right) \geq \tau\right) \tag{4}$$

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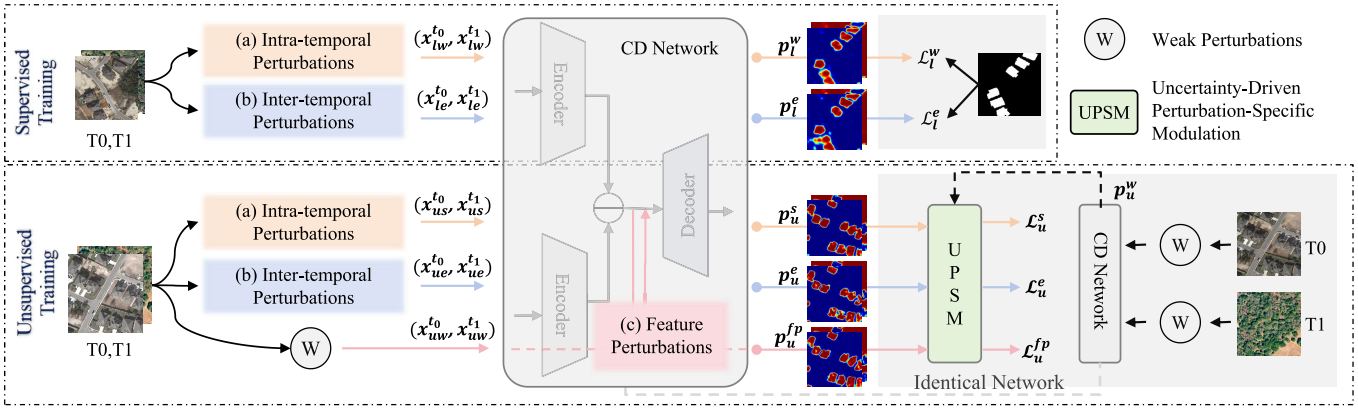


Fig. 2. Flowchart of the proposed UA-ECL for SSCD. Predictions from the multiperturbation branch align with the ground truth on labeled data and with the weakly perturbed stream's predictions on unlabeled data. Perturbation-specific pixel-level uncertainty maps are generated for each unlabeled stream to quantitatively assess the reliability of supervision signals and adaptively guide the learning process.

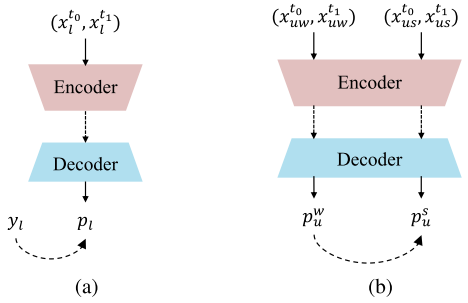


Fig. 3. Diagram of the weak-to-strong consistency learning for CD. (a) Supervised. (b) Unsupervised.

$$\mathcal{L}_u = \frac{1}{|B_u|} \sum \mathbf{M} \cdot \mathrm{CE}\left(\mathbf{p}_u^s, \mathbf{p}_u^w\right). \tag{5}$$

To ensure stable training during the initial stage, τ is applied to filter out ambiguous pixels and retain reliable regions, $\mathbf{M}$, for unsupervised learning, thereby reducing interference from noisy pseudo-labels and facilitating discriminative feature extraction.

However, given the complexity of imaging and environmental conditions during RSI acquisition, relying solely on intratemporal perturbations is insufficient to cope with the diverse nonsemantic variations across temporal observations. To overcome this limitation, we extend the consistency learning paradigm for CD by employing multiple parallel perturbation branches that explicitly model intertemporal, intratemporal, and feature-level variations. Furthermore, we proposed an ECL strategy to jointly optimize labeled and unlabeled data by enforcing consistency across predictions from multibranch perturbations.

B. ECL for RSI Pairs

As a critical paradigm in semi-supervised learning, the weak-to-strong consistency framework benefits from incorporating strong perturbations [32], [50], [57], [58], which are essential for enhancing representation learning. As shown in Fig. 4(a), existing perturbation strategies typically treat each temporal image independently, rather than exploiting

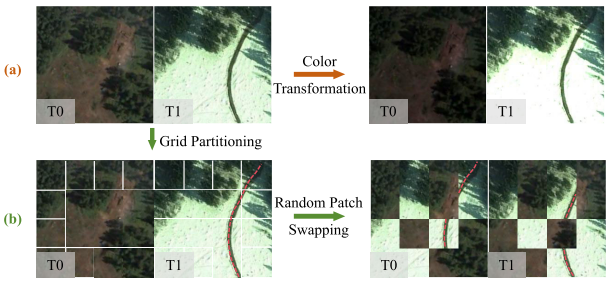


Fig. 4. intratemporal and intertemporal perturbations. (a) intratemporal perturbations (color transformation and CutMix) are applied independently to each temporal image. (b) Intertemporal perturbation (SwapMix) exchanges corresponding regions between bitemporal RSIs without altering the locations of semantic changes.

the bitemporal relationships to extract temporally consistent semantic change representations. Due to differences in the acquisition times of bitemporal images, diverse imaging and environmental conditions often lead to intraclass discrepancies that are potentially misinterpreted as land surface changes. Furthermore, these discrepancies often mislead CD networks to focus on irrelevant information during feature extraction rather than on the discriminative features essential for CD. These limitations tend to be more pronounced in scenarios with limited labeled data, ultimately constraining network generalization performance.

To this end, to enhance the robustness of CD networks against pseudo-changes, we aim to strengthen the perception of intertemporal information within a single temporal feature-extraction process, thereby mitigating the influence of irrelevant global seasonal and radiometric information on CD. Therefore, a novel intertemporal perturbation technique $\mathcal{A}^e$, SwapMix, is proposed to produce cross-temporal images with mixed environmental variations. Specifically, a proportion of nonoverlapping patches between the two temporal images was exchanged, as detailed in Algorithm 1. As illustrated in Fig. 4(b), the locations of the changes in the SwapMixed RSI pair remained unaffected, while the local radiometric and structural information were perturbed. A robust CD network is expected to produce consistent change predictions under

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Algorithm 1 SwapMix: Intertemporal Patch Exchange

Require: Bi-temporal images $\mathbf{x}^{t_0}, \mathbf{x}^{t_1} \in \mathbb{R}^{H\times W\times C}$, patch size k

Generate a random binary mask $\mathbf{I} \in \{0,1\}^{\frac{H}{k}\times\frac{W}{k}}$ with $\mathbf{I}_{i,j} \sim$ Bernoulli(0.5)

Upsample $\mathbf{I}$ to pixel-level mask $\hat{\mathbf{I}} \in \{0,1\}^{H\times W}$ via nearest-neighbor interpolation, and expand it along the channel dimension

$\mathbf{x}_e^{t_0} \leftarrow \hat{\mathbf{I}} \odot \mathbf{x}^{t_1} + (1-\hat{\mathbf{I}}) \odot \mathbf{x}^{t_0}$

$\mathbf{x}_e^{t_1} \leftarrow \hat{\mathbf{I}} \odot \mathbf{x}^{t_0} + (1-\hat{\mathbf{I}}) \odot \mathbf{x}^{t_1}$

return $\mathbf{x}_e^{t_0}, \mathbf{x}_e^{t_1}$

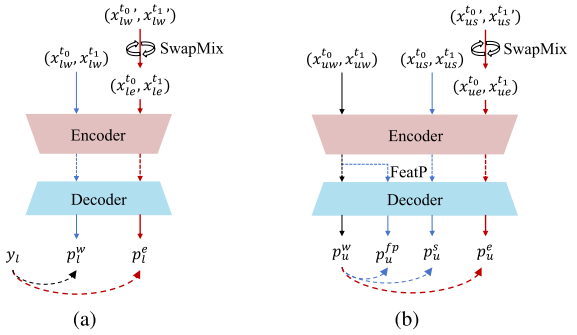


Fig. 5. Proposed ECL strategy consists of (a) SECL strategy and (b) UECL strategy. SwapMix performs random patch exchange between RSI pairs, and FeatP applies feature perturbation to the difference features.

both radiometrically consistent and inconsistent conditions. Consequently, we propose ECL for both supervised and unsupervised branches to enforce consistency between the predictions obtained with and without SwapMix perturbation. Notably, introducing ECL into the supervised process can facilitate robust unsupervised training. This mechanism encourages the network to learn perturbation-invariant semantic representations while reducing sensitivity to nonsemantic variations, thereby alleviating pseudo-change interference. However, due to unreliable predictions and perturbation noise, directly imposing consistency constraints between them could introduce instability during early training. Therefore, we employ unified and reliable targets to supervise each perturbed branch separately, using pseudo-labels generated from weakly perturbed data for the unsupervised branch and ground truth for the supervised branch, thereby implicitly promoting consistency across branches.

1) Unsupervised ECL (UECL): In the unsupervised branch, intertemporal perturbation $\mathcal{A}^e$ is applied to strongly perturbed RSI pairs $(\mathbf{x}_{us}^{t_0\prime}, \mathbf{x}_{us}^{t_1\prime})$ to further exchange cross-temporal information for semantically change-awareness perception enhancement of the network under multiple disturbances

$$\left(\mathbf{x}_{ue}^{t_0}, \mathbf{x}_{ue}^{t_1}\right) = \mathcal{A}^e\left(\mathbf{x}_{us}^{t_0\prime}, \mathbf{x}_{us}^{t_1\prime}\right). \tag{6}$$

Following that, the predictions $\mathbf{p}_u^e$ are obtained from inputs $(\mathbf{x}_{ue}^{t_0}, \mathbf{x}_{ue}^{t_1})$. As shown in Fig. 5(b), the UECL constraint is applied among $\mathbf{p}_u^e$, and predictions from two auxiliary parallel streams without SwapMix, including intratemporal perturbed results $\mathbf{p}_u^s$ and feature-level perturbed results $\mathbf{p}_u^{fp}$. Specifically, as in (3), intratemporal perturbation applies image-level strong augmentations $\mathcal{A}^s$ to both bitemporal images, such as color transformations, Gaussian blur, and CutMix. In addition, feature-level perturbation (FeatP) is implemented via a dropout operator $\mathcal{P}$ on the bitemporal difference deep features $\mathbf{f}_d$

$$\mathbf{f}_d^w = \left|\mathcal{F}\left(\mathbf{x}_{uw}^{t_0}\right) - \mathcal{F}\left(\mathbf{x}_{uw}^{t_1}\right)\right| \tag{7}$$

$$\mathbf{p}_u^{fp} = \mathcal{G}\left(\mathcal{P}\left(\mathbf{f}_d^w\right)\right). \tag{8}$$

By aligning predictions across these diverse perturbations, the CD network is encouraged to leverage unlabeled data to fully enrich feature representations. Specifically, the weakly perturbed predictions $\mathbf{p}_u^w$, filtered by a confidence threshold τ, are employed as pseudo-labels to calculate the cross-entropy losses with the predictions $\mathbf{p}_u^e$, $\mathbf{p}_u^s$, and $\mathbf{p}_u^{fp}$, which are then aggregated to constitute the total unsupervised loss $\mathcal{L}_u$

$$\mathcal{L}_u = \frac{1}{|B_u|}\sum \mathbf{M}\cdot\left(\mathrm{CE}\left(\mathbf{p}_u^e, \mathbf{p}_u^w\right) + \mathrm{CE}\left(\mathbf{p}_u^s, \mathbf{p}_u^w\right) + \mathrm{CE}\left(\mathbf{p}_u^{fp}, \mathbf{p}_u^w\right)\right). \tag{9}$$

2) Supervised ECL (SECL): To achieve more stable and consistent training, ECL is also adopted into the supervised training process. The weakly perturbed labeled RSI pairs $(\mathbf{x}_{lw}^{t_0\prime}, \mathbf{x}_{lw}^{t_1\prime})$ are processed by intertemporal perturbation for SECL

$$\left(\mathbf{x}_{le}^{t_0}, \mathbf{x}_{le}^{t_1}\right) = \mathcal{A}^e\left(\mathbf{x}_{lw}^{t_0\prime}, \mathbf{x}_{lw}^{t_1\prime}\right). \tag{10}$$

As shown in Fig. 5(a), both predictions $\mathbf{p}_l^e$ and $\mathbf{p}_l^w$ are supervised by the ground truth $\mathbf{y}_l$ in SECL to guide the network to learn semantically discriminative representations from SwapMix-enhanced pairs

$$\mathcal{L}_s = \frac{1}{|B_l|}\sum\left(\mathrm{CE}\left(\mathbf{p}_l^w, \mathbf{y}_l\right) + \mu_1\mathrm{CE}\left(\mathbf{p}_l^e, \mathbf{y}_l\right)\right) \tag{11}$$

where μ_1 is the weighting parameters. This enables the network to establish a robust preliminary perception of cross-temporal correlations, facilitating unsupervised consistency learning.

Through joint training of UECL and SECL, we establish a unified perturbation-consistency principle across the labeled and unlabeled branches that integrates intertemporal, intratemporal, and feature-level perturbations. By leveraging hierarchical feature learning with ECL constraints, the CD network progressively focuses on discriminative land-cover semantic properties across radiometric and seasonal variations rather than on irrelevant background information, thereby alleviating incorrect segmentation of pseudo-changes.

C. Uncertainty-Driven Perturbation-Specific Modulation

In contrast to the supervised loss computed from ground truth, the unsupervised loss $\mathcal{L}_u$ lacks sufficient reliability due to uncertainty in the supervisory signals. The uncertainty usually originates from actual semantic changes in small objects, intricate change boundaries, and confounding factors such as radiometric inconsistencies and phenological variations associated with seasonal variation. Therefore, the consistency learning in these pseudo-change and challenging regions may lead the CD network into unreliable training. However, a fixed confidence threshold for filtering reliable pseudo-labels is inflexible, hindering the balance between pseudo-label reliability and the effective utilization of unlabeled data. As

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a result, even after filtering by the confidence threshold, erroneous supervision information may remain in the pseudo-labels obtained from weakly perturbed RSI pairs.

To address this limitation, it is critical to estimate uncertainty maps adaptively within a single forward pass. Since regions with noisy pseudo-labels often exhibit ambiguous land-cover characteristics, they are more susceptible to non-semantic perturbations, leading to inconsistent predictions across perturbed inputs. Consequently, predicted probabilities under different perturbations tend to diverge substantially in pseudo-change and semantically ambiguous areas, yet remain generally similar in regions with stable land-cover properties. To exploit this property, we design a UPSM mechanism to generate independent uncertainty maps for the three perturbation streams, enabling adaptive modulation during unsupervised training. These uncertainty maps adaptively attenuate the contribution of unreliable pseudo-label regions during multiperturbation consistency learning, thereby yielding more robust supervision. Leveraging the sensitivity of KL divergence to relative discrepancies between probability distributions, we employ it as the difference metric to derive a pixel-level uncertainty map $\mathbf{u}^*$ for the prediction $\mathbf{p}_u^*$

$$\mathbf{u}^* = \sum_{c=0}^{1} \text{softmax}\left(\mathbf{p}_u^w(c)\right) log \frac{\text{softmax}\left(\mathbf{p}_u^w(c)\right)}{\text{softmax}\left(\mathbf{p}_u^*(c)\right)} \tag{12}$$

where c denotes the class type, and $\mathbf{p}_u^*$ refers to $\mathbf{p}_u^e$, $\mathbf{p}_u^s$, or $\mathbf{p}_u^{fp}$. Following that, the obtained uncertainty maps are utilized to adaptively emphasize regions with more confident predictions in the loss function while reducing the weights of potentially noisy labels for unsupervised training

$$\mathcal{L}_u^* = \frac{1}{|B_u|}\left(\sum \mathbf{M} \cdot \frac{\text{CE}\left(\mathbf{p}_u^*, \mathbf{p}_u^w\right)}{\exp\left(\mathbf{u}^*\right)} + \sum \mathbf{M} \cdot \mathbf{u}^*\right). \tag{13}$$

The first term enforces consistency over low-uncertainty regions while relaxing this constraint in high-uncertainty areas. The second term regularizes the uncertainty map to prevent large uncertainty estimates, thereby avoiding insufficient optimization during unsupervised training. By integrating threshold filtering with soft uncertainty penalization, the proposed strategy adaptively modulates supervision signals for unlabeled data and stabilizes the training process. Finally, the three perturbation-specific uncertainty maps $\mathbf{u}^e$, $\mathbf{u}^s$, and $\mathbf{u}^{fp}$ are computed independently for each stream and applied separately to modulate their respective loss terms, thereby providing each perturbation branch with tailored supervision signals.

D. Multibranch Semi-Supervised Loss Function

For labeled data, the supervised loss $\mathcal{L}_s$ is defined as the combination of cross-entropy losses between the ground truth and the predictions from the weakly and intertemporal perturbed inputs. For unlabeled data, the weakly perturbed prediction $\mathbf{p}_u^w$ is used as a pseudo-label to supervise three types of predictions generated from perturbed branches: the intertemporal perturbed prediction $\mathbf{p}_u^e$, the intratemporal perturbed prediction $\mathbf{p}_u^s$, and the feature-level perturbed prediction $\mathbf{p}_u^{fp}$. Subsequently, the UPSM mechanism adaptively adjusts the three cross-entropy loss terms, yielding three robust unsupervised losses $\mathcal{L}_u^e$, $\mathcal{L}_u^s$, and $\mathcal{L}_u^{fp}$. Therefore, the final unsupervised loss is formed as follows:

$$\mathcal{L}_u' = \mu_2 \mathcal{L}_u^{fp} + \mu_3 \mathcal{L}_u^s + \mu_4 \mathcal{L}_u^e \tag{14}$$

where μ_2, μ_3, and μ_4 are the weighting parameters. Ultimately, the SSCD network is jointly trained with reliable annotated subsets and abundant unlabeled data to enhance its generalization capability by

$$\mathcal{L} = \mathcal{L}_s + \mathcal{L}_u'. \tag{15}$$

IV. Experiments

A. Datasets

We evaluated the effectiveness of our approach through experiments on three publicly available datasets.

The LEVIR-CD [59] dataset consists of 637 pairs of 0.5-m resolution bitemporal RSIs from Google Earth, captured between 2002 and 2018. Each image has a spatial size of 1024 × 1024 pixels and covers urban areas from 20 cities in Texas, USA. This dataset focuses on building-related changes, including villa residences, tall apartments, small garages, and large warehouses. A large number of tiles consisting solely of negative samples makes this benchmark valuable for evaluating models' ability in application performance. For experiments, the images were cropped into 256 × 256 patches and split into 7120 pairs for training, 1024 pairs for validation, and 2048 pairs for testing.

The SVCD [60] dataset was acquired from Google Earth and includes seven pairs of high-resolution images, each 4725 × 2700 pixels, and four pairs with minor seasonal variation, each 1900 × 1000 pixels. These were cropped into 16 000 image pairs of 256 × 256 pixels, with spatial resolutions ranging from 0.03 to 1m, and divided into 10 000 pairs for training, 3000 for validation, and 3000 for testing. The significant seasonal variations between image pairs enable us to evaluate how a CD approach filters out irrelevant changes caused by environmental conditions.

SYSU-CD [61] dataset comprises 20 000 pairs of 256 × 256 aerial image patches with 0.5-m resolution, collected in Hong Kong between 2007 and 2014. This dataset captures diverse change information, including urban development, vegetation shifts, road expansion, and coastal construction. These patches are derived from 800 large-scale image pairs, and were split into 6:2:2 for training, testing, and validation. Compared with existing CD datasets, SYSU-CD exhibits greater diversity in change types, serving as a valuable benchmark for deep-learning-based CD in densely built, rapidly evolving urban environments.

B. Evaluation Criteria

To evaluate the performance of the proposed method, we employ four widely used metrics in CD, including $F1$-score (F1), changed-class intersection over union (IoU_c), kappa coefficient (Kappa), and overall accuracy (OA). These metrics are computed based on four fundamental quantities: true

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positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). Among the four metrics, $\mathrm{IoU_c}$ explicitly quantifies the overlap of changed regions, whereas the remaining three indices are composite measures used to evaluate the model's overall performance. Their formulas are as follows:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Pre} \times \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}}, \quad \mathrm{IoU_c} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{16}$$

$$\mathrm{OA} = \frac{\mathrm{TN} + \mathrm{TP}}{\mathrm{TN} + \mathrm{FP} + \mathrm{TP} + \mathrm{FN}}, \quad \mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{P_e}}{1 - \mathrm{P_e}} \tag{17}$$

where Pre and Rec denote precision and recall, which are defined as follows:

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{18}$$

The $\mathrm{P_e}$ represents the proportion of expected agreement

$$\mathrm{P_e} = \frac{(\mathrm{FP} + \mathrm{TP}) \cdot (\mathrm{FN} + \mathrm{TP}) + (\mathrm{FP} + \mathrm{TN}) \cdot (\mathrm{FN} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{19}$$

The ranges of F1, $\mathrm{IoU_c}$, and OA span from 0% to 100%, and Kappa takes values between −1 and 1. Higher values of the four metrics indicate better performance.

C. Implementation Details

To enable a fair comparison with prior work [62], a widely used network architecture, DeepLabv3+ with a ResNet-50 backbone, is adopted as the segmentation model. All experiments are conducted in the PyTorch framework on an NVIDIA GeForce RTX 4090 with 24-GB memory. The model is trained for 80 epochs on each dataset using SGD as the optimizer, with an initial learning rate of 0.02 and polynomial decay. Each mini-batch comprises six labeled and six unlabeled image pairs. Weak perturbations randomly resize input images by a factor of 0.5–2.0, followed by random cropping and horizontal flipping. For the other three parallel perturbation strategies, the detailed operations are as follows.

1) *intratemporal Perturbations:* Weak perturbations are applied in the supervised branch, while color jittering, Gaussian blur, and CutMix are further applied in the unsupervised branch with probabilities of 0.8, 0.5, and 0.5, respectively.
2) *Intertemporal Perturbations:* SwapMix is applied after weak augmentation in the supervised branch and after intratemporal perturbations in the unsupervised branch.
3) *Feature Perturbations*: A 50% channel dropout is applied to the difference features between the difference features and the decoder.

All auxiliary branch loss weights, μ_1, μ_2, μ_3, and μ_4, are set to 0.5. SwapMix partitions the images into nonoverlapping 64×64 patches on a uniform grid, and randomly selects 50% of them for exchange between the two temporal images at each training iteration. Pseudo-labels with confidence scores above 0.9 for LEVIR-CD, SVCD, and SYSU-CD are retained as reliable regions. Each experiment is repeated five times, and the reported results are the average of the five runs.

D. Detection Performance Comparison

We compare our UA-ECL strategy with six state-of-the-art SSCD methods to demonstrate its superiority. Including SemiCD [32], FPA [31], PUF [33], ECPS [45], Unimatch [34], and Unimatch V2 [57] on the LEVIR-CD, SVCD, and SYSU-CD datasets. ECPS adopts a pseudo-labeling paradigm via cross-model supervision, and the remaining approaches focus on designing consistency regularization schemes tailored to CD. To ensure a fair comparison, all methods are trained on identical data splits with 1%, 5%, 10%, and 20% labeled samples, and share the same input resolution, batch size, and training schedule. Since SemiCD and FPA involve specific feature-level designs or loss functions, we retain their original ResNet-50 backbone with dilated convolutions, while other methods use a standard ResNet-50. In addition, we incorporated a supervised-only baseline (Sup-base), trained solely on the four corresponding labeled splits, and a fully supervised model trained on the complete dataset for comprehensive comparison.

As shown in Table I, the proposed UA-ECL consistently outperforms competing methods in most cases. On the building-change dataset LEVIR-CD, UA-ECL achieves a remarkable performance improvement over other methods in $\mathrm{IoU_c}$ with only 1% labeled data. At the same time, under the 5%, 10%, and 20% settings, it surpasses the second-optimal method by 1.16%, 0.70%, and 0.82%, respectively. Since LEVIR-CD focuses solely on building changes, models can achieve superior performance with limited supervision compared with those trained on datasets containing more diverse change types. Therefore, our ablation study employs a 1% labeled split of LEVIR-CD and a 5% labeled split of SVCD to validate the effectiveness of our method with less labeled data. It is also worth noting that the performance gap between UA-ECL and competing methods narrows as the amount of labeled data increases, a commonly observed trend in semi-supervised learning. As more labeled data becomes available, ground-truth supervision increasingly dominates the optimization, reducing the relative contribution of consistency regularization. Nevertheless, UA-ECL consistently outperforms all competing methods across all label settings, and its advantages are particularly significant in low-label regimes. On multitype CD datasets, UA-ECL consistently maintains superior performance on both SVCD and SYSU-CD, demonstrating robustness to substantial style variation and complex urban scenes. In addition, UA-ECL achieves performance comparable to or superior to that of the fully supervised approach on LEVIR-CD and SYSU-CD at 10% and 20% labeled ratios. Overall, these experimental results indicate that the joint effect of the proposed ECL and UPSM modules exploits latent semantic information from unlabeled data, thereby substantially enhancing CD performance in complex scenarios.

Unimatch, PUF, and ECPS achieve competitive performance across several experimental settings. Nonetheless, the lack of intertemporal relationship modeling in these methods limits interaction between bitemporal images, thereby compromising the ability to address pseudo-changes caused by radiometric differences and seasonal variations. ECPS achieves

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT SSCD METHODS ON DIFFERENT TRAINING RATIOS ON LEVIR-CD, SVCD, AND SYSU-CD DATASETS

Ratio	Method	LEVIR-CD				SVCD				SYSU-CD			
		F1(%)	IoUc(%)	Kappa	OA(%)	F1(%)	IoUc(%)	Kappa	OA(%)	F1(%)	IoUc(%)	Kappa	OA(%)
1%	SemiCD	72.95	57.43	0.7167	97.52	62.88	45.88	0.5824	91.59	69.60	53.39	0.6040	85.87
	FPA	70.47	54.40	0.6914	97.40	63.62	46.66	0.5845	90.93	74.28	59.10	0.6632	87.82
	PUF	84.37	73.06	0.8362	98.56	68.39	51.97	0.6454	93.08	**78.41**	**64.49**	**0.7201**	**90.11**
	ECPS	73.07	57.57	0.7168	97.35	57.18	40.04	0.5306	91.77	59.72	43.72	0.4107	67.89
	Unimatch	83.02	70.98	0.8221	98.44	71.31	55.44	0.6791	**93.85**	77.71	63.55	0.7127	89.98
	UnimatchV2	86.35	76.00	0.8568	98.72	68.19	51.77	0.6428	92.97	75.63	60.84	0.6889	89.36
	Sup-base	66.68	50.02	0.6499	96.77	58.96	41.80	0.5452	91.74	71.69	55.89	0.6294	86.62
	UA-ECL	**87.96**	**78.54**	**0.8735**	**98.84**	**72.06**	**56.33**	**0.6848**	93.64	76.99	62.59	0.7033	89.65
5%	SemiCD	84.23	72.75	0.8344	98.49	80.06	66.75	0.7738	95.25	76.99	62.62	0.7009	89.37
	FPA	83.55	71.66	0.8273	98.42	82.04	69.55	0.7963	95.72	79.94	66.59	0.7383	90.62
	PUF	89.77	81.45	0.8925	99.00	85.56	74.77	0.8370	96.70	79.25	65.63	0.7329	90.72
	ECPS	88.32	79.08	0.8770	98.83	81.36	68.61	0.7889	95.62	77.73	63.57	0.7012	88.59
	Unimatch	89.52	81.03	0.8898	98.97	86.81	76.70	0.8517	97.06	79.57	66.07	0.7392	91.09
	UnimatchV2	89.29	80.65	0.8874	98.96	86.52	76.24	0.8481	96.96	79.37	65.81	0.7359	90.92
	Sup-base	83.66	71.92	0.8285	98.45	73.75	58.44	0.7062	94.36	76.13	61.46	0.6873	88.70
	UA-ECL	**90.48**	**82.61**	**0.8998**	**99.06**	**87.59**	**77.92**	**0.8600**	**97.18**	**80.22**	**66.97**	**0.7468**	**91.30**
10%	SemiCD	86.19	75.74	0.8547	98.63	85.00	73.91	0.8293	96.35	80.67	67.61	0.7516	91.39
	FPA	86.31	75.28	0.8560	98.65	85.74	75.04	0.8377	96.53	79.70	66.25	0.7373	90.77
	PUF	90.37	82.43	0.8987	99.05	89.21	80.52	0.8778	97.48	80.67	67.61	0.7516	91.39
	ECPS	89.87	81.61	0.8934	98.98	90.19	82.13	0.8881	97.58	**82.44**	**70.12**	**0.7695**	91.63
	Unimatch	90.04	81.89	0.8953	99.02	90.00	81.83	0.8873	97.73	80.33	67.13	0.7487	91.39
	UnimatchV2	90.11	82.00	0.8960	99.03	89.98	81.79	0.8870	97.72	79.69	66.24	0.7401	91.07
	Sup-base	87.07	77.11	0.8641	98.73	79.43	65.90	0.7694	95.52	78.14	64.12	0.7163	89.96
	UA-ECL	**90.79**	**83.13**	**0.9031**	**99.08**	**91.55**	**84.43**	**0.9046**	**98.06**	81.72	69.10	0.7638	**91.72**
20%	SemiCD	86.56	76.31	0.8586	98.67	87.96	78.51	0.8626	97.02	81.47	68.74	0.7614	91.70
	FPA	86.31	75.91	0.8560	98.65	87.95	78.49	0.8625	97.02	81.56	68.87	0.7600	91.44
	PUF	90.44	82.55	0.8994	99.06	92.38	85.84	0.9138	98.22	81.06	68.16	0.7564	91.54
	ECPS	90.40	82.49	0.8990	99.04	**94.70**	**89.93**	**0.9394**	**98.68**	82.03	69.54	0.7641	91.43
	Unimatch	90.24	82.21	0.8973	99.03	93.34	87.52	0.9247	98.46	80.27	67.05	0.7483	91.41
	UnimatchV2	90.13	82.03	0.8962	99.03	93.11	87.11	0.9222	98.41	79.98	66.65	0.7429	91.10
	Sup-base	88.48	79.33	0.8788	98.87	84.99	73.91	0.8313	96.68	79.30	65.70	0.7307	90.42
	UA-ECL	**90.93**	**83.37**	**0.9045**	**99.10**	94.11	88.87	0.9333	98.63	**82.94**	**70.85**	**0.7798**	**92.30**
100%	Sup-base	90.81	83.17	0.9033	99.09	95.87	92.08	0.9532	99.03	81.73	69.11	0.7649	91.83

competitive CD results on several datasets, but its effectiveness is highly limited by the quality of pseudo-labels generated under insufficient supervision. On the SVCD dataset, ECPS achieves the lowest performance with a 1% labeled split, yet the highest with a 20% labeled split, indicating substantial instability in its predictions. In contrast, UA-ECL remains particularly robust and stable under all experimental conditions.

E. Visualization Results

To visually demonstrate the superiority of the proposed UA-ECL framework against other competing approaches, some segmentation results from the LEVIR-CD with 1% labeled splits, SVCD with 5% labeled splits, and SYSU-CD with 5% labeled splits are presented in Fig. 6. The visualizations are ordered from top to bottom, showing the test results of all compared methods for each dataset.

Rows 1–3 illustrate several building regions with subtle changes that are radiometrically similar to unchanged areas, making it difficult to accurately segment the change regions. In contrast, UA-ECL can detect almost all such change regions while obtaining few erroneous predictions. By performing patch-level cutting and mixing, SwapMix introduced finer spatial structures and more complex environmental variants, thereby enabling the CD network to distinguish subtle changes from unchanged regions. Rows 4–6 present CD results from different methods on the SVCD dataset. Substantial pseudo-changes caused by seasonal variations and objects with complex structures pose significant challenges for distinguishing the semantic changes of interest. With the help of ECL strategies and UPSM, UA-ECL achieves more precise boundaries and significantly fewer FPs in complex scenarios, including snow-covered road sections and vegetation influenced by seasonal dynamics. Furthermore, as shown in

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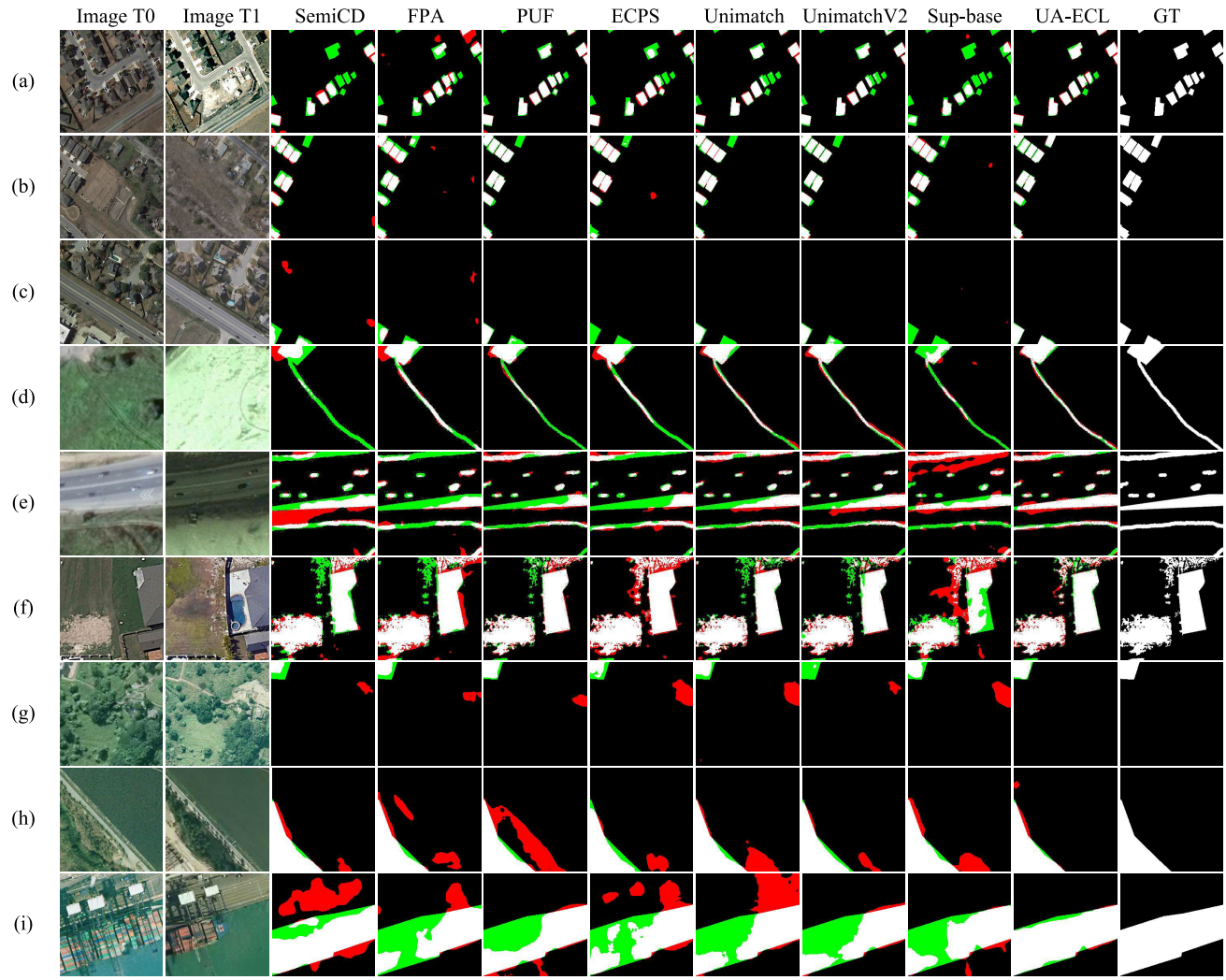


Fig. 6. Some CD results obtained by different SSCD methods, where white indicates TP, black indicates TN, red indicates FP, and green indicates FN. (a)–(c), (d)–(f), and (g)–(i) Results on the test sets of different methods trained on the LEVIR-CD, SVCD, and SYSU-CD datasets, respectively.

rows 7–9, UA-ECL maintains robust performance in urban scenes of SYSU-CD, highlighting its generalization across diverse CD applications.

Furthermore, to demonstrate the robustness of UA-ECL in semantic feature extraction, we visualize the difference feature maps between bitemporal images on the SVCD dataset, which exhibits significant seasonal variations. We present several challenging examples where snow cover, vegetation growth, and varying illumination conditions lead to pronounced radiometric discrepancies across the same land surface. As shown in Fig. 7, the high-response regions generated by UA-ECL largely align with the ground-truth change regions, indicating that the model can effectively suppress irrelevant interference in bitemporal images and learn stable feature representations. However, other methods produce the difference feature maps that respond not only to true change regions but also to irrelevant radiometric variations. Robust feature representations provide a reliable foundation for subsequent change decoding, enabling the model to better distinguish true semantic changes from pseudo-variations. These visualizations demonstrate the effectiveness of the proposed method in suppressing non-semantic disturbances and enhancing robustness in complex real-world scenarios.

F. Ablation Studies

The proposed UA-ECL framework comprises ECL strategies driven by UECL and SECL in both supervised and unsupervised branches, along with a UPSM mechanism that facilitates reliable consistency learning through uncertainty estimation. To verify the effectiveness of each component, ablation studies were conducted on the LEVIR-CD with 1% labeled samples and the SVCD with 5% labeled samples.

1) *Component 1*: Supervised training performed solely on labeled data.
2) *Component 2*: Extending Component 1 by incorporating UECL to enable unsupervised training with unlabeled data.
3) *Component 3*: Replacing the supervised baseline in Component 2 with SECL to strengthen the supervised learning process.
4) *Component 4*: Building upon Component 3 by introducing UPSM into consistency learning (UA-ECL).

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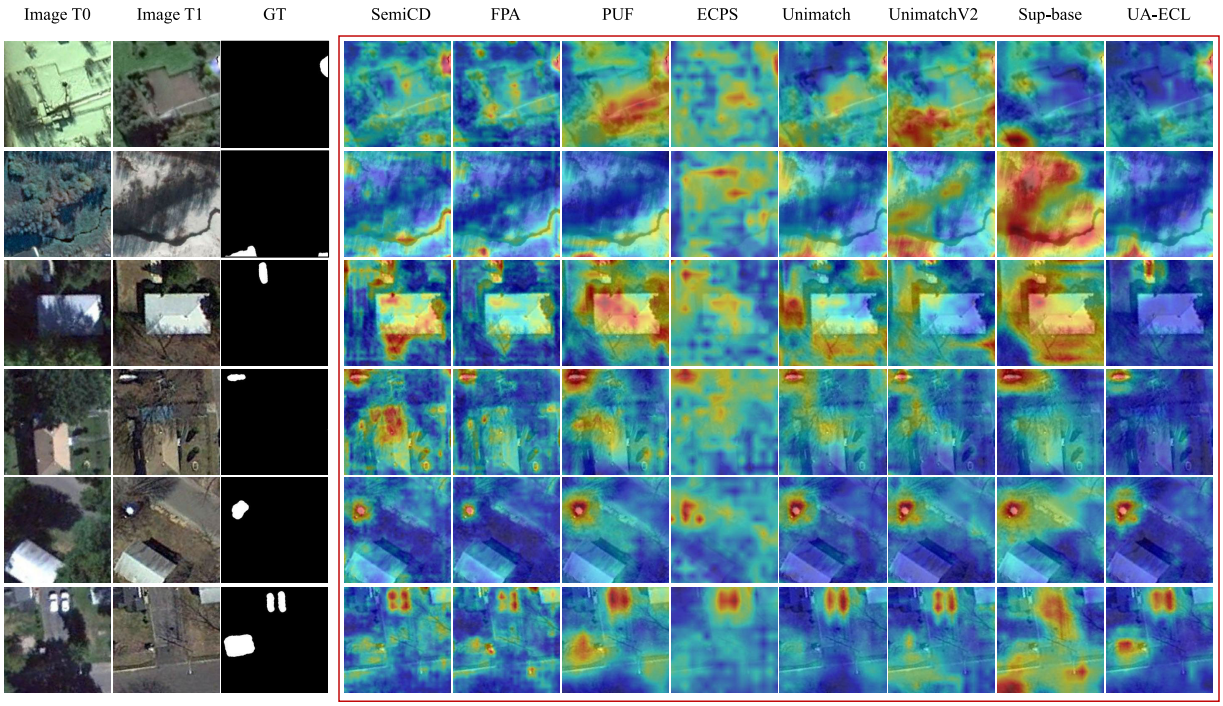


Fig. 7. Visualization of the difference feature maps between bitemporal images on the SVCD dataset under the 5% labeled setting.

TABLE II
EFFECTIVENESS OF EACH COMPONENT ON LEVIR-CD AND SVCD DATASET. THE ◊ IS MEASURED AGAINST THE BASELINE

Method	Sup-base	UECL	SECL	UPSM	LEVIR-CD(1%)				SVCD(5%)			
					F1(%)	IoUc(%)	Kappa	OA(%)	F1(%)	IoUc(%)	Kappa	OA(%)
Component 1	✓				66.68	50.02	0.6499	96.77	58.96	41.80	0.5452	94.36
Component 2	✓	✓			84.03	72.47	0.8326	98.51	86.95	76.92	0.8535	97.13
Component 3		✓	✓		84.64	73.40	0.8390	98.57	87.25	77.38	0.8565	97.15
Component 4		✓	✓	✓	87.96	78.54	0.8735	98.84	87.59	77.92	0.8600	97.18
Gain ◊					↑ 21.28	↑ 28.52	↑ 0.2236	↑ 2.07	↑ 28.63	↑ 36.12	↑ 0.3148	↑ 2.82

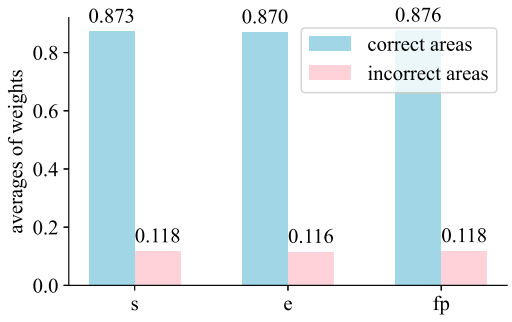


Fig. 8. Average weights $1/\exp(\mathbf{u}^*)$ for correctly and incorrectly predicted pseudo-label regions on the LEVIR-CD dataset. The "s," "e," and "fp" represent the streams for $\mathbf{p}_u^s$, $\mathbf{p}_u^e$, and $\mathbf{p}_u^{fp}$, respectively.

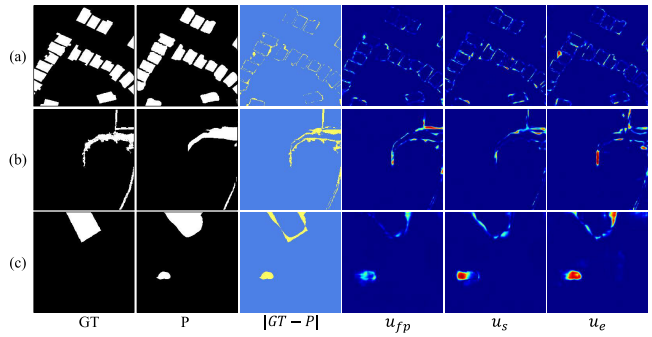


Fig. 9. Visualization of uncertainty maps generated by UPSM. (a)–(c) Samples of LEVIR-CD, SVCD, and SYSU-CD datasets, respectively. "P" indicates the pseudo-label originated from $\mathbf{p}_u^w$.

As shown in Table II, the quantitative results indicate that the full UA-ECL learning strategy achieves the best performance. The baseline component, trained solely on supervised signals, achieves limited performance with F_1 of 66.68% and 58.96% on the LEVIR-CD and SVCD datasets, respectively. By incorporating UECL, the CD network benefits from additional change information from unlabeled data, where consistency regularization enforces agreement under both intratemporaland intertemporal perturbations, thereby increasing the $F1$-score to 84.03% and 86.95%. By further embedding the SECL into the supervised training branch,

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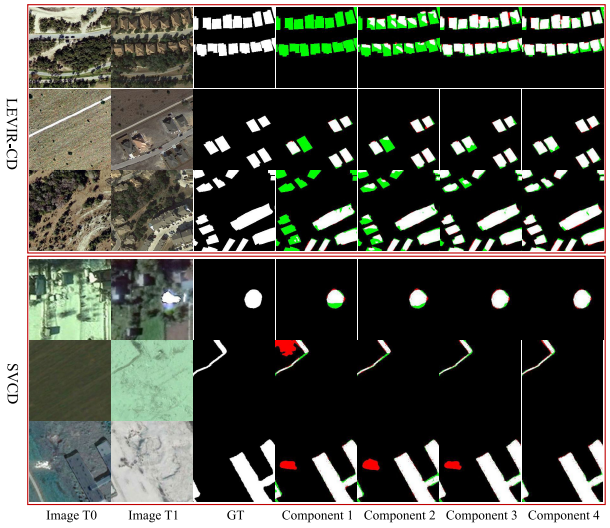


Fig. 10. Qualitative visualization under different ablation settings.

the network yields additional gains, with $F1$-score increasing to 84.64% and 87.25%. To obtain more reliable supervision from unlabeled data, we integrate UPSM to perform pixel-wise weighted updates on the loss terms. By comparing the performance of Components 3 and 4, the incorporation of the UPSM strategy achieves improvements of 3.32% and 0.34% in the $F1$-score on the LEVIR-CD and SVCD datasets, respectively. As illustrated in Fig. 8, correctly predicted regions in the pseudo-labels are assigned higher weights, indicating that uncertainty estimation effectively leverages reliable information while suppressing noise of pseudo-labels, thereby contributing to more stable training. Furthermore, Fig. 9 visualizes uncertainty maps generated by UPSM, and the third column shows the binary difference map from pseudo-labels and ground truth, with highlighted areas indicating incorrect predictions. High-uncertainty regions largely overlap with pseudo-change and ambiguous areas, further indicating the ability of our proposed method to suppress interference information for CD. Furthermore, Fig. 10 presents qualitative results under different ablation settings, where the predictions progressively improve as additional components are incorporated.

In summary, the proposed UA-ECL framework demonstrates strong capability to leverage unlabeled data, achieving $F1$-score improvements of 21.28% and 28.63% on the two benchmark datasets compared with training schemes that rely solely on labeled samples.

In addition, our UA-ECL framework proposes a novel intertemporal perturbation technique that performs multiple random patch swaps to enhance the perception of cross-temporal discrepancy. To verify its effectiveness, we develop a comparative method, Component 2-wo SwapMix, by removing SwapMix from Component 2 and retaining three perturbation streams. As shown in Table III, improvements in $F1$-scores across all labeled-split settings when incorporating SwapMix demonstrate its effectiveness on bitemporal representation learning for change information perception.

TABLE III
COMPARISON OF $F1$-SCORES WITH AND WITHOUT SWAPMIX ON TWO DATASETS ACROSS THE FOUR SPLIT SETTINGS

Dataset	Method	1%	5%	10%	20%
LEVIR-CD	Component 2-wo SwapMix	83.02	89.53	90.02	90.21
	Component 2	84.03	90.26	90.61	90.85
	Gain ◊	↑ 1.01	↑ 0.73	↑ 0.59	↑ 0.64
SVCD	Component 2-wo SwapMix	71.18	86.81	89.68	93.29
	Component 2	72.11	86.95	90.74	93.45
	Gain ◊	↑ 0.93	↑ 0.14	↑ 1.06	↑ 0.16

TABLE IV
COMPUTATIONAL COST COMPARISON ON SVCD DATASET WITH 5% LABELED DATA

Method	Train		Reference	
	Time (min) ↓	GPU (GB) ↓	GFLOPs ↓	F1(%) ↑
SemiCD	8.3	10.01	73.23	80.06
FPA	5.2	11.65	73.23	82.04
PUF	12.1	19.70	29.10	85.56
ECPS	15.0	18.47	29.10	81.36
Unimatch	5.5	10.15	29.10	86.81
Unimatch V2	4.1	7.31	29.10	86.52
UA-ECL	6.6	12.13	29.10	87.59

G. Computational Cost Comparison

In this section, we analyze the computational cost of UA-ECL during training and inference, and compare it against competing methods on the SVCD dataset with a 5% labeled split. Specifically, we report the average training time per epoch (Time), the peak memory usage during training, and the model's computational complexity (GFLOPs), summarized in Table IV.

It should be noted that SemiCD and FPA use a dilated ResNet-50 backbone, which retains higher resolution feature maps by replacing strided convolutions with dilated ones in later stages, resulting in substantially higher GFLOPs (73.23) than other methods (29.1). ECPS and PUF incur notably higher GPU memory consumption, as ECPS relies on a multimodel ensemble, while PUF introduces an auxiliary training branch. In contrast, UA-ECL maintains a moderate GPU memory consumption while achieving the highest $F1$-score of 87.59%, surpassing the second-best method, Unimatch, by 0.78%. Furthermore, despite achieving superior performance, our method maintains competitive training time and GPU memory consumption compared with existing semi-supervised methods. These results demonstrate that UA-ECL achieves a favorable tradeoff between computational overhead and detection performance, making it a practical and effective choice for SSCD in real-world scenarios.

H. Parameter Analysis

For a more comprehensive analysis, the sensitivity of the proposed UA-ECL framework to critical hyperparameters is discussed in detail, including the size of randomly swapped patches and the SwapMix embedding location. In addition,

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TABLE V

PERFORMANCE OF UA-ECL WITH DIFFERENT SWAPMIX PATCH SIZES ON THE LEVIR-CD AND SVCD DATASET WITH 5% LABELED DATA

Patch Size	LEVIR-CD			SVCD		
	F1(%)	IoUc(%)	OA(%)	F1(%)	IoUc(%)	OA(%)
64×64	**90.48**	**82.61**	**99.06**	87.59	77.92	97.18
32×32	90.32	82.35	99.04	87.62	77.97	97.18
16×16	90.46	82.58	99.05	**87.71**	**78.12**	**97.20**
8×8	90.28	82.29	99.04	87.43	77.67	97.14

TABLE VI

IMPACT OF SWAPMIX EXCHANGE RATIO ON THE LEVIR-CD AND SVCD DATASETS WITH 5% LABELED DATA

Exchange Ratio	LEVIR-CD			SVCD		
	F1(%)	IoUc(%)	Kappa	F1(%)	IoUc(%)	Kappa
10%	90.22	82.19	0.8971	86.95	76.91	0.8526
20%	90.05	81.89	0.8953	87.34	77.53	0.8571
30%	90.34	82.38	0.8984	87.52	77.81	**0.8606**
40%	90.42	82.51	0.8991	87.50	77.78	0.8574
50%	**90.48**	**82.61**	**0.8998**	**87.59**	**77.92**	0.8600

as a key hyperparameter for initially filtering reliable regions in pseudo-labels, the threshold τ can help stabilize the semi-supervised learning and also needs to be analyzed in detail.

1) Impact of Patch Size on SwapMix: By exchanging bitemporal patches, UA-ECL facilitates effective fusion of temporal environmental information and enhances spatial structural diversity. To investigate the impact of patch size in SwapMix, we evaluate the model across four configurations on the LEVIR-CD and SVCD datasets. As shown in Table V, UA-ECL achieves optimal performance with a patch size of 64 × 64 in the LEVIR-CD dataset and 16 × 16 in the SVCD dataset, surpassing the worst-performing configuration (8 × 8) by 0.2% and 0.28% in $F1$-score, respectively. When the patch size is reduced to 8 × 8, the swap operation becomes overly disruptive to the spatial continuity of ground objects, thereby impairing the model's ability to capture contextual semantic information. In addition, the performance with patch sizes of 64 × 64 and 32 × 32 is comparable to that of 16 × 16, and all are substantially better than the 8 × 8 setting. These results indicate that the model performance becomes relatively insensitive to further size variations within the range of 16 × 16 to 64 × 64. Therefore, a larger patch size of 64 × 64 is adopted in our final implementation to preserve the structural continuity of the mixed RSIs and achieve stable, reliable segmentation performance.

2) Exchange Ratio of SwapMix: The exchange ratio controls the proportion of spatial regions swapped between bitemporal images, thereby directly affecting the degree of intertemporal perturbation. To investigate the influence of the exchange ratio in SwapMix, we conduct parameter analysis experiments by varying the proportion of swapped regions from 10% to 50% on both LEVIR-CD and SVCD datasets. As shown in Table VI, performance gains are limited at smaller exchange ratios, such as 10% and 20%, suggesting that limited intertemporal interaction cannot fully exploit the potential of ECL. As the exchange ratio increases, performance gradually improves. This phenomenon indicates that sufficient intertemporal interaction is essential for robust cross-temporal land-cover representations to be learned through ECL. When the exchange ratio reaches 50%, the model achieves the best performance on both LEVIR-CD and SVCD datasets. Therefore, we adopt 50% as the default setting in all experiments.

TABLE VII

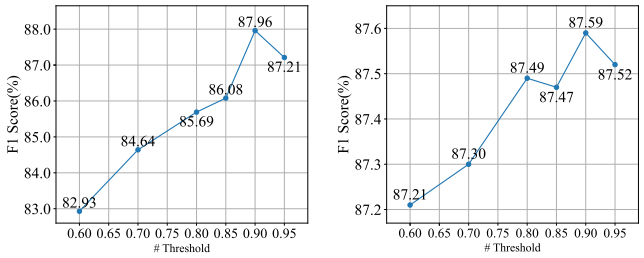
PERFORMANCE OF THE UA-ECL (WITH FSWAPMIX) STRATEGY ON THE LEVIR-CD AND SVCD DATASETS

Ratio	LEVIR-CD			SVCD		
	F1(%)	IoUc(%)	Kappa	F1(%)	IoUc(%)	Kappa
1%	87.83	78.31	0.8722	71.45	55.58	0.6778
5%	90.00	81.82	0.8948	87.79	78.24	0.8621
10%	90.65	82.90	0.9016	91.29	83.98	0.9015
20%	90.67	82.94	0.9019	94.11	88.88	0.9333

3) SwapMix at Feature Level: To further assess whether intertemporal perturbation generalizes from the image level to the feature level, we apply FSwapMix, a feature-level variant of SwapMix. The intertemporal exchange operation is transferred to the feature space for both the supervised and unsupervised branches, while all other perturbation streams remain unchanged. Specifically, FSwapMix is applied to the feature representations extracted after 4× downsampling in the Siamese encoder. To maintain a spatial granularity comparable to the 64 × 64 configuration used in the image domain, the bitemporal feature maps are partitioned into nonoverlapping 16 × 16 patches, and 50% of the patches are randomly selected and exchanged between the two temporal features. As shown in Table VII, although UA-ECL with FSwapMix exhibits slightly lower performance than UA-ECL with SwapMix, it remains strong competitiveness relative to the methods listed in Table I. Across the four split settings of the LEVIR-CD dataset, UA-ECL with FSwapMix outperforms the suboptimal approach by 1.48%, 0.23%, 0.28%, and 0.23% in $F1$-score, respectively. These results demonstrate that SwapMix in the image or feature domain can enhance the representational capacity by leveraging cross-temporal interactions.

4) Threshold τ Settings for Reliable Pseudo-Label Generation: Due to limited labeled data availability, confirmation bias and perturbation-induced noise may introduce errors into pseudo-labels for unlabeled samples. To ensure stable training during the early stages, we retain a confidence threshold to mitigate the impact of noisy supervision. To evaluate the effect of the threshold τ on model performance, we conducted experiments with varying thresholds on the LEVIR-CD and SVCD datasets. As shown in Fig. 11, as the threshold increases, CD performance on both datasets exhibits an initial improvement followed by a subsequent degradation. This trend indicates that a suitable confidence threshold stabilizes the initial training process and integrates effectively with UPSM's adaptive weighting mechanism, thereby further stabilizing training. Specifically, a low threshold introduces numerous unreliable supervisory signals, and an excessively high thresh-

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(a) (b)

Fig. 11. Performance comparison with different thresholds τ trained on LEVIR-CD and SVCD with 1% and 5% labeled ratio. (a) 1% labeled LEVIR-CD. (b) 5% labeled SVCD.

old reduces the number of available training pixels, leading to sparse supervision and unstable optimization. Overall, under limited annotations, UA-ECL employs a fixed threshold to suppress noise and utilizes UPSM for fine-grained supervision refinement, thereby achieving stable model performance.

V. Conclusion

In this work, we investigate perturbation strategies explicitly designed for CD within a consistency regularization framework to mitigate the adverse effects of radiometric discrepancies on the effective utilization of unlabeled RSI pairs. Due to the diverse representations of RSIs in real-world scenarios caused by complex imaging and environmental conditions, we integrate intertemporal, intratemporal, and feature-level perturbations within a unified framework to broaden the perturbation space under limited labeled data, organizing them into parallel streams for perturbation-specific consistency learning. Specifically, we propose a novel intertemporal perturbation technique, SwapMix, to perceive cross-temporal discrepancies and diversify the spatial structure of the RSIs. On this basis, we propose an ECL strategy to enhance the robustness of feature representations by aligning predictions across streams under multiple perturbation types for labeled and unlabeled data. Furthermore, the proposed UPSM mechanism generates pixel-level uncertainty to evaluate the reliability of supervision signals and adaptively regulate the training process. Benefiting from consistent learning across multibranch perturbations with fine-grained supervision modulation, UA-ECL achieves precise boundary segmentation and robust SSCD in complex scenarios involving subtle changes and ambiguous change regions. Our future work will explore vision foundation models for leveraging unlabeled data and their deployment in large-scale CD applications, including nonagricultural monitoring.

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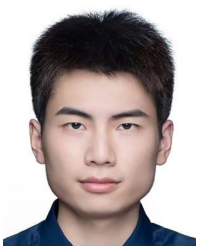
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